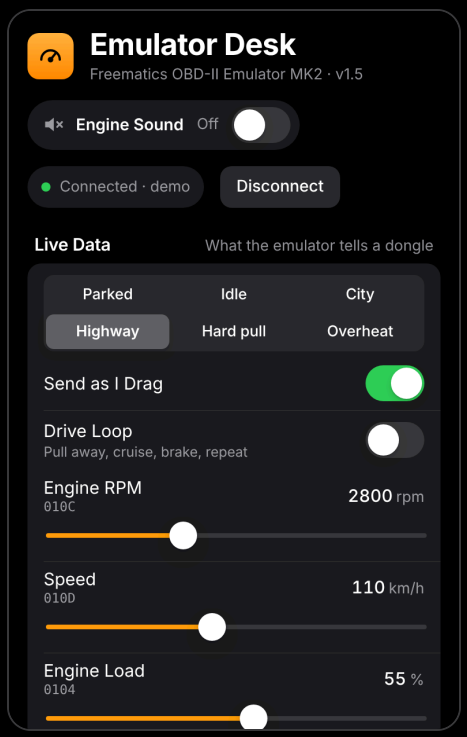
Emulator Desk

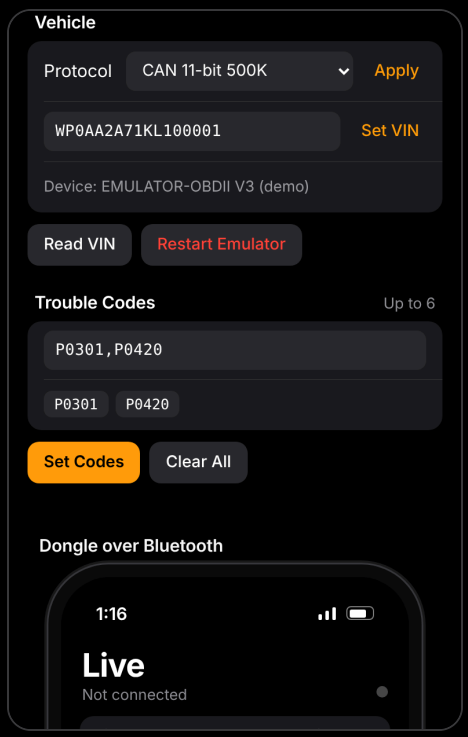
A Mac control panel for the Freematics OBD-II Emulator MK2

App **v1.5**
Document **v1.5**

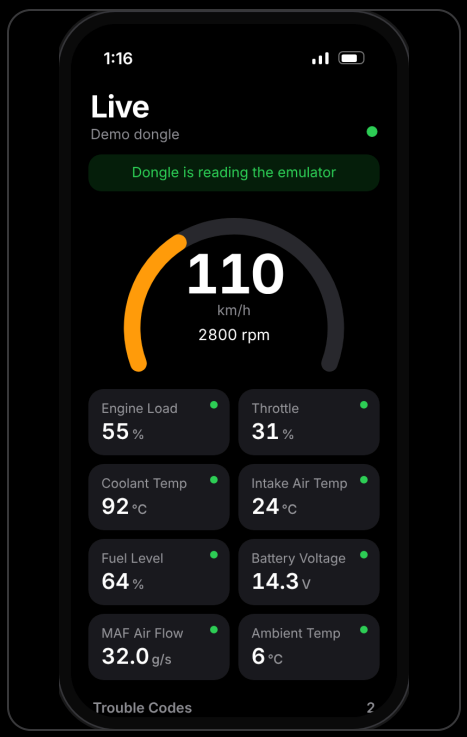
Emulator Desk drives a Freematics MK2 emulator from Google Chrome over its USB cable, so a Mac can do what the Windows-only Freematics GUI does. You set live engine values, the VIN, the bus protocol and trouble codes, and any OBD-II dongle or scan tool plugged into the emulator reads them as if it were in a real car. It runs from a web page, with nothing to install, and follows Apple's Human Interface Guidelines in light and dark mode.



Live data
Ten sliders for the common Mode 01 values, with presets from Parked to Overheat. Values go out as you drag.



Vehicle and codes
A compact panel for protocol, VIN and up to six trouble codes, set or cleared in one tap.



Dongle phone view
A phone-style screen showing everything a Bluetooth dongle reads: gauge, ten values, codes, VIN and protocol.

10 live PIDs with sliders	**6** driving presets plus a drive loop	**7** bus protocols to choose from	**1** link to open it, nothing to install

- **Live data**

Ten common values sent as raw OBD-II bytes. Drive Loop cycles pulling away, cruising and braking.

- **Dongle over Bluetooth**

Connect a Bluetooth LE ELM327 dongle and a phone-style screen shows everything it reads. Dots show whether each value matches the emulator.

- **Good to know**

On connect the sliders read back what the emulator is serving. Reset (ATZ) restarts it in a few seconds and zeroes every value. Its EMULATOR-OBDII V3 firmware can set the VIN but not read it back, and it can't tell whether a dongle is plugged in.

- **Engine sound**

Flip the Engine Sound switch and a flat-six plays, built from single cylinder firings. It thumps at idle, howls near redline and crackles on lift-off.

Technical setup

Sources, integrations and what runs where

App **v1.5**
Document **v1.5**

Sources and libraries

Piece	Version	What it does	Loaded from	Licence
Emulator Desk	1.5	The whole app: UI, command queue, PID encoding, presets, drive loop, engine sound, dongle check	`index.html` on GitHub Pages	Written for this app
Freematics AT command set	EMULATOR-OBDII V3 (tested)	Commands the app sends: `ATE0`, `ATZ`, `ATSET`, `ATGET` (PIDs), `ATCLR DTC`	Built into the emulator	Freematics
OBD-II PID formulas	SAE J1979	Byte encodings for each slider (e.g. RPM × 4 as two bytes, temperature + 40)	Hard-coded in the app	Public standard
System fonts	n/a	SF Pro and SF Mono on Mac, Arial and Consolas elsewhere	The operating system	OS fonts

No third-party libraries, CDNs or web fonts are used.

Integrations

Integration	Used for	Details
Web Serial API	Talking to the emulator	`navigator.serial.requestPort()`, opened at 38400 baud, 8N1. Commands end in `\r`. You pick the port in Chrome's chooser; no auth. Works in Chrome and Edge, not Safari or Firefox.
Freematics OBD-II Emulator MK2	The device being controlled	USB to serial through a CP2102 (newer units, `/dev/cu.usbserial-*`) or CH340 bridge, both with drivers built into macOS. Needs its 12V adapter for power. The app labels the chip from the USB vendor ID.
Web Bluetooth	Reading a plugged-in dongle	`navigator.bluetooth.requestDevice()`, you pick the dongle. Finds a notify and a write characteristic (FFE0, FFF0, 18F0 and similar), sends ELM327 commands (`ATZ`, `ATSP0`, all ten Mode 01 PIDs, `03`, `0902`, `ATRV`, `ATDP`). Locked dongles such as Automatic's don't answer.
Web Audio API	Engine sound	An AudioWorklet fires six cylinders every two revolutions (RPM ÷ 120 × 6 pulses a second), each a decaying thump plus combustion noise, through two exhaust pipe resonators, a muffler low-pass and a soft clipper, then EQ and a compressor. Starts only when you flip the switch.
GitHub Pages	Hosting	`davidanthonyalleyne-dot.github.io/emulator-desk`, served from the `main` branch of the public `emulator-desk` repo. Static files only.

What runs where

On your Mac: everything. The page runs in Chrome and sends one command at a time, waiting up to 1.2 seconds for OK or ERROR before the next.

Network: one page load from GitHub Pages. After that the app makes no web requests.

Data kept: nothing. Slider values, VIN and codes live only in the open page and on the emulator itself.

Demo mode: add `?demo` to the address to try the app without hardware. The screenshots in this document use it with sample values.

Regenerating this document

In `docs/overview`, run `node shoot.js` to retake the screenshots in demo mode at 390 × 844, then `node pdf.js` to print `overview.pdf` on Letter pages.